



Digital Transformations for
Supporting Next-Generation Labour

Co-funded by the
Erasmus+ Programme
of the European Union



MetaLab

MacPaw AI Lab in VRChat for the NEXT-Study Metaverse

Nomination I — Metaverse representation of the university

Team MetaLab

Igor Sikorsky Kyiv Polytechnic Institute

Team members

Sviatoslav Pavlenko, Denys Iliencko, Dmytro Poshytyniuk,
Mark Mamenko, Kateryna Shozda

nextstudy.eu

Sviatoslav Pavlenko

Igor Sikorsky KPI

Role in the project

Team Lead. Coordination, task split, deadlines and communication with the consultant.

Denys Iliencko

Igor Sikorsky KPI

Role in the project

3D Artist. Room and equipment modelling, retopology, mesh prep for texturing.

Dmytro Poshytyniuk

Igor Sikorsky KPI

Role in the project

Look Development. PBR textures and Unity materials for realistic surfaces.

Mark Mamenko

Igor Sikorsky KPI

Role in the project

UdonSharp Developer. Interactive terminals, object state sync across players.

Kateryna Shozda

Igor Sikorsky KPI

Role in the project

QA / VR Tester. In-headset checks, FPS, collisions and network stability.

How we work

Agile team of five. Modelling, texturing, C# logic and Unity assembly run in parallel, synced over Git. Regular VR playtests.

THE PROBLEM

No shared room in remote study

MacPaw AI Lab · NEXT-Study Metaverse

MetaLab brings MacPaw AI Lab into VRChat. Seminar rows, boards and workstations live in one multiplayer hall so students can practise together with a sense of real presence.

PROJECT GOALS

- Visual accuracy: real proportions of furniture and gear
- Functional space for SE and networking practice
- Extensible layout for later AI / simulators
- Stable FPS via retopo, PBR and light baking
- Multiplayer VRChat world for NEXT-Study

Selected Location

NEXT

MacPaw AI Lab

The world rebuilds MacPaw AI Lab: seminar rows, critique boards, a lounge and workstations. Visitors share one room instead of a video grid.



The space in VRChat

NEXT

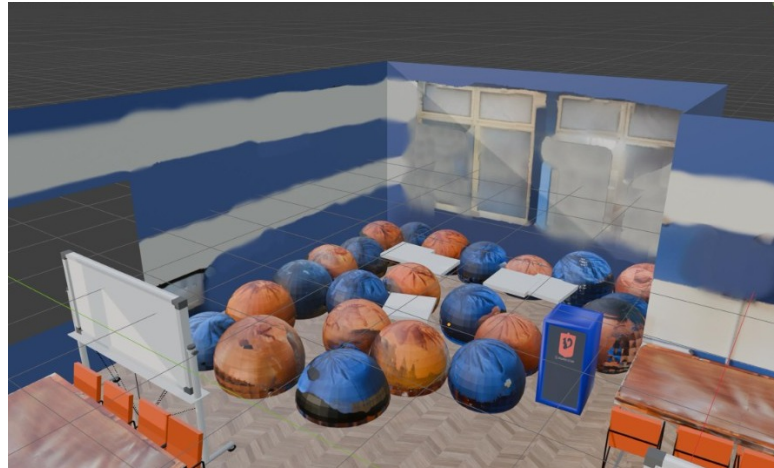


Classroom layout



Optimised room mesh with baked lighting, PBR materials and collisions for VRChat comfort.

Lounge and boards



Seminar tables, whiteboards and a rest zone for critiques and informal work.

Workstations



Interactive ecosystem: terminals, switches, markers — UdonSharp syncs state between users.

Tools and software

Unity (scene, collisions, light baking) · VRChat SDK3 / Udon · UdonSharp (C#) · Blender · Adobe Substance 3D Painter (PBR) · Git.

Development process

1) Block out MacPaw AI Lab. 2) Model and retopo furniture and equipment. 3) Paint PBR materials. 4) Bake lighting and collisions in Unity. 5) Add UdonSharp interaction and publish the world.

VRChat integration

SDK3 Worlds: spawns, audio zones, synced object state. Published world wrld_b1c73436-f022-4f98-9172-671f9f0da989. Prefer PC / PCVR.

Technical challenges

Polygon and material budgets vs visual fidelity. Baking instead of realtime lights. Keeping FPS high enough to avoid VR sickness. Networking interactive props without breaking the instance.

Educational use

In NEXT-Study Metaverse a group meets in the same virtual hall for remote practicals, hackathons and applicant tours — with the spatial feel of a real classroom.

User experience

Spawn, walk the aisles, sit at a workstation, use boards and equipment, see classmates in the same room. Presence replaces a flat video call.

Further development

Hook AI mentors and network simulators through OSC and HTTP, refine Quest performance, keep adding equipment as modular prefabs.

Demo and Project Materials

NEXT

Project demo

https://vrchat.com/home/world/wrld_b1c73436-f022-4f98-9172-671f9f0da989/info

Project materials

GitHub: github.com/mamenkomarkus-creator/MetaLab-NEXT

Docs: [docs/README.md](#) · [docs/CONCEPT.md](#) · [docs/VRCHAT.md](#)

World ID: wrld_b1c73436-f022-4f98-9172-671f9f0da989

Additional resources

VRChat: hello.vrchat.com NEXT: nextstudy.eu



GitHub



VRChat



NEXT

Digital Transformations for
Supporting Next-Generation Labour

Thank you!
Questions?

